

Implementation Plan: WhatsApp Notification Migration

Project Goal: Phased replacement of loan email notifications with a resilient WhatsApp pipeline.

1. Project Guardrails

In Scope

- **Infrastructure:** Notification queue, retry logic, and provider adapters.
- **Observability:** Status tracking (Sent/Delivered/Read) and Admin logs.
- **Triggers:** Full loan lifecycle automation.
- **Operations:** Rollback triggers and feature flag controls.

Out of Scope (Phase 1)

- Two-way conversational AI/Chatbots.
- Customer support ticketing via WhatsApp.

2. Execution Roadmap

Estimated Duration: 18–29 Working Days

Phase 0: Strategy & Compliance (Days 2–4)

- **Tasks:** Finalize provider (Meta vs. Twilio), approve message templates, and define legal consent UX.
- **Key Decision:** Confirming the source of truth for phone numbers (User.telephone vs. verified WhatsApp field).
- **Deliverables:** Provider Decision Note, Template Catalog v1.

Phase 1: Core Foundation (Days 3–5)

- **Tasks:** Prisma DB schema migrations for outbox/delivery logs; scaffolding the notification service in backend/src/lib/.
- **Logic:** Implement worker loop, retry backoff, and global feature flags.
- **Deliverables:** DB Migration script, Unit-tested worker module.

Phase 2: Integration & Webhooks (Days 4–6)

- **Tasks:** Build provider adapter and secure webhook endpoints for status callbacks.
- **Security:** Implement signature verification for incoming provider data.
- **Deliverables:** Validated send-path in Sandbox, Persistent delivery state tracking.

Phase 3: Event Wiring (Days 3–4)

- **Tasks:** Connect domain events (Submit, Approve, Disburse, Repay, Complete, Overdue) to the notification engine.
- **Deliverables:** Event-to-Template mapping, E2E trigger tests.

Phase 4: UI & Admin Experience (Days 2–3)

- **Tasks:** Frontend consent capture (opt-in + timestamp) and Admin visibility dashboard.
- **Deliverables:** Consent UX components, Admin observability page.

Phase 5: Controlled Rollout (Days 4–7)

- **Tasks:** Internal pilot → 10% Production cohort → Monitoring → 100% Ramp-up.
- **Deliverables:** Pilot Report, Final KPI Analysis.

3. Risk Management & Continuity

Risk Register

Risk	Impact	Mitigation Strategy
Template Rejection	High	Submit for approval during Phase 0; draft fallback versions.
Invalid Formatting	Med	Enforce strict E.164 normalization on the frontend/API.
Compliance Gap	High	Audit log for every opt-in; mandatory consent timestamps.
Duplicate Delivery	Low	Implement Idempotency Keys per loan event.

Rollback Protocol

1. **Trigger:** Critical delivery failure or data corruption.
2. **Action:** Toggle WHATSAPP_ENABLED=false via feature flag.
3. **Fallback:** System automatically reverts to Email transport for high-priority notifications.
4. **Recovery:** Drain the queue or replay failed events once the provider is stable.

4. Technical Quality Standards

Testing Matrix

- **Unit:** Template rendering logic, backoff algorithms, and phone normalization.
- **Integration:** Webhook signature validation and DB state transitions (queued → delivered).
- **UAT:** Verifying "No Consent = No Message" and verifying trigger accuracy.

Monitoring & KPIs

- **Primary Metrics:** Send Success Rate (>95%), Delivery/Read rates.
- **Alerting:** Trigger notifications if API 5xx errors exceed 5% over 15 minutes.

5. Stakeholder Check-In

Immediate actions required for kick-off:

1. **Approval:** Select provider (Meta vs. Twilio) and budget.
2. **Copy-writing:** Approve text for the 6 core loan event templates.

3. **Legal:** Sign-off on the consent wording and UI placement.
4. **Resource:** Confirm Backend/Frontend engineer availability for the 4-week window.